
Law & Data Science

Laws, Treaties & Sources — Reference

Master's Degree in Data Science
Department of Mathematics — School of Science

Academic Year 2024–25

Scope: every legal instrument cited in the course — international instruments, Council of Europe conventions, EU primary law (treaties + Charter), EU secondary law, the 2020 Data Strategy package, and leading case law. Loci (articles) and dates are reported. Brief notes give just enough content to place each instrument; conceptual development lives in the companion summary.

Contents

1	International Instruments	3
1.1	UN Universal Declaration of Human Rights (UDHR) — 1948	3
1.2	International Covenant on Civil and Political Rights (ICCPR) — 1966	3
1.3	UN Convention on the Rights of the Child — 1989	3
1.4	OECD Privacy Guidelines — 1980 (Paris)	3
2	Council of Europe (CoE) Instruments	3
2.1	European Convention on Human Rights (ECHR) — 1950	3
2.2	Convention 108 — 28 January 1981 (“Data Privacy Day”)	3
2.3	Convention 108+ — 18 May 2018	4
3	EU Primary Law	4
3.1	Founding & Amending Treaties (timeline)	4
3.2	Treaty on European Union (TEU) — key articles	4
3.3	Treaty on the Functioning of the EU (TFEU) — key articles	5
3.4	Charter of Fundamental Rights of the EU (CFR)	5
3.5	General Principles of EU Law	6
4	EU Secondary Law	6
4.1	Typical & Atypical Acts (Art. 288 TFEU)	6
4.2	Data-Protection Directives	6
4.3	Data-Protection Regulations	7
4.4	GDPR — Regulation (EU) 2016/679: key articles	7
5	2020 European Data Strategy — Legislative Package	8
6	Leading Case Law	8

1 International Instruments

Sources outside the EU/CoE legal orders. Mostly **soft law** (morally/politically binding, not enforceable before a court) with two notable hard-law exceptions.

1.1 UN Universal Declaration of Human Rights (UDHR) — 1948

- Adopted by the **UN General Assembly**, 1948; **soft law** (no binding force).
- **Art. 12** — first recognition of privacy: protection against arbitrary interference with privacy, family, home, correspondence, and attacks on honour/reputation. A **negative**, liberty-based notion of privacy.

1.2 International Covenant on Civil and Political Rights (ICCPR) — 1966

- First time privacy is recognised as a fundamental right in **hard law**: a convention signed and ratified by many states, entering into force in their jurisdictions.
- (Source labels it “International Covenant on Civil and Legal Rights”).)

1.3 UN Convention on the Rights of the Child — 1989

- International convention, **binding** on ratifying states (entered into force once enough states ratified).
- **Art. 16** — no child shall be subjected to arbitrary interference with privacy; children directly protected. Dignity-oriented protection of children specifically.

1.4 OECD Privacy Guidelines — 1980 (Paris)

- **Soft law**; OECD = Organisation for Economic Cooperation and Development.
- **First international instrument** to set principles for processing personal data; universal standards balancing data-subject rights against controllers’ needs.
- **Eight principles**: collection limitation; data quality; purpose specification; use limitation; security safeguards; openness; individual participation; accountability. (These principles are the ancestors of the GDPR principles.)

2 Council of Europe (CoE) Instruments

The CoE is a separate legal order from the EU (nowadays **46 member states**; Russia expelled). Its court is the **European Court of Human Rights (ECtHR)**, Strasbourg, competent only over the ECHR.

2.1 European Convention on Human Rights (ECHR) — 1950

- Completed 1950; sets out rights, procedures and institutions (incl. the ECtHR).
- **Art. 8** — Right to respect for private and family life: no interference by a public authority **except** as provided by law and necessary in a democratic society (national security, public safety, prevention of crime, protection of health/rights of others).
- Used by the ECtHR even recently to rule on privacy *and* on unlawful processing of personal data.

2.2 Convention 108 — 28 January 1981 (“Data Privacy Day”)

- Convention for the protection of individuals w.r.t. **automated processing of personal data**.
- **First binding international instrument** on personal data protection (binding for contracting states); open to signature beyond the CoE.
- Recalls the OECD principles; covers **public and private** sectors; sets the basis for trans-border data flows and for processing in police/enforcement contexts.

- Enforcement **cannot** be judged by the ECtHR (its jurisdiction is limited to the ECHR). Source of the **accountability** principle later in the GDPR.

2.3 Convention 108+ — 18 May 2018

- Modernisation of Convention 108 by its contracting states; standards evolve with technology, addressing emerging/emergency privacy challenges.
- Date is significant: adopted 18 May 2018, days before GDPR application (25 May 2018); the two orders aligned. Core principles: protection against personal-data abuses; regulation of transborder data flows; fair & lawful collection; legitimate purposes; purpose-bound processing; storage duration; data quality (minimisation/proportionality); sensitive data; right to know & rectify; possible overriding interests.

3 EU Primary Law

Top of the EU hierarchy of sources. Comprises the founding/amending **treaties**, the **Charter of Fundamental Rights**, and **general principles** developed by the Court of Justice. Cannot be contradicted by secondary or supplementary law.

3.1 Founding & Amending Treaties (timeline)

Treaty	Year	Significance
Treaty of Rome	1957	Created the European Economic Community (EEC); now the TFEU (effective 1958).
Single European Act	1986	Amending treaty; advanced the single market.
Treaty of Maastricht	1992	Established the EU ; now the TEU (effective 1993). From 1992 the EU is a “union”, not a mere community.
Treaty of Amsterdam	1996/97	Institutional amendments (e.g. size of the Parliament) for enlargement.
Treaty of Nice	2001	Prepared the 2004 enlargement (15→25 members); drafted the first Charter of Fundamental Rights as soft law (“Nice Charter”).
Treaty of Lisbon	2007	Effective 2009 ; gave the Charter the same legal value as the treaties (hard law); amended the TEU/TFEU as they stand today.
Accession treaties	—	Terms/conditions for new members; adjust existing treaties.

The EU rests on **two founding treaties**: the **TEU** (objectives, values, institutions — effective 1993, Maastricht) and the **TFEU** (detailed organisational/functional provisions — effective 1958, Rome).

3.2 Treaty on European Union (TEU) — key articles

- **Art. 2 — Values** of the Union: human dignity, freedom, democracy, equality, the rule of law, respect for human rights (incl. minorities). Pre-condition for membership.
- **Art. 3 — Objectives**: peace and well-being; area of freedom & free movement; internal market & sustainable development; economic & monetary union (the euro); upholding values/interests externally; pursued within conferred competencies (hence not all law is harmonised).
- **Art. 6(3)** — fundamental rights (incl. from the ECHR and constitutional traditions) are general principles of EU law; basis for protecting fundamental rights.

- **Art. 13** — establishes the **EU institutions**.
- **Art. 39** — data protection in the area of **Common Foreign & Security Policy (CFSP)**; here rules are adopted by the **Council alone** (not the Parliament). Special law prevailing over Art. 16 TFEU in its domain.
- **Art. 49** — **accession procedure**: any European state respecting Art. 2 values may apply; involves all institutions except the Court of Justice; Council acts by majority; conditions amend the treaties.

3.3 Treaty on the Functioning of the EU (TFEU) — key articles

- **Art. 16 / 16(1)** — everyone has the **right to the protection of personal data**; legal basis for EU data-protection legislation (Parliament + Council, ordinary procedure). Personal data protection becomes part of primary law; lower sources cannot contradict it.
- **Art. 216** — the Union may conclude **international agreements** to achieve treaty objectives; once concluded they bind the EU and its member states (part of the *acquis*).
- **Art. 217** — **association agreements** with third countries / international organisations (reciprocal rights and obligations).
- **Art. 218** — **procedure** for concluding international agreements (Court of Justice may rule on compatibility).
- **Art. 263** — **direct appeals / action for annulment** of an EU act (lack of competence, infringement, misuse of power); filed within **2 months + 10 days**.
- **Art. 265** — **action for failure to act** by an EU institution (preceded by a letter of formal notice; 2 months to respond).
- **Art. 267** — **preliminary rulings**: a national court refers a question of EU-law interpretation/validity; the CJEU says *how* to interpret, the national judge decides the case. Binding on the referring court.
- **Art. 288** — defines the **typical acts** of secondary law (regulations, directives, decisions, recommendations, opinions). See Section 4.
- **Art. 340(2)** — **action for damages**: non-contractual liability of the EU for harm caused by its institutions.

3.4 Charter of Fundamental Rights of the EU (CFR)

- Proclaimed **2000** (Nice, soft law); became **legally binding in 2009** with the Treaty of Lisbon — same legal status as the treaties. Acts as a quasi-constitution providing a hard core of fundamental rights.
- Origin: customs recognised by the Court of Justice in the 1960s to safeguard fundamental rights.
- **Seven chapters**: (1) Dignity — inviolable; (2) Freedoms — incl. private/family life and protection of personal data; (3) Equality; (4) Solidarity — social/workers' rights; (5) Citizens' rights — e.g. vote in EU elections; (6) Justice — access to justice, fair trial; (7) General provisions — safeguard clauses.

Key CFR articles:

- **Art. 1** — human dignity (inviolable).
- **Art. 7** — right to **private and family life** (protection *against interference*).
- **Art. 8** — autonomous **right to protection of personal data**: §1 everyone has the right; §2 fair processing, for specified purposes, on consent or another legitimate basis, with rights of access & rectification; §3 control by an **independent authority**. Art. 8 governs the *conditions/rules* of processing (distinct from Art. 7's anti-interference focus). The GDPR

finds its legal basis here.

- **Art. 41** — (per source) basis cited for the **free economic initiative** underpinning the Data Strategy. (*Note: the freedom to conduct a business is usually Art. 16 CFR; the source cites Art. 41.*)
- **Art. 52** — **Safeguard clause / limitations**: any limitation on Charter rights must be **provided by law**, respect the **essence** of the right, pursue legitimate general-interest objectives, and be **necessary and proportionate**. Basis for balancing conflicting rights.

3.5 General Principles of EU Law

Developed by the Court of Justice, later codified; constantly evolving from the constitutional traditions of member states:

- Legal certainty; legitimate expectation; **primacy** of EU law; **direct effect** of EU law; proportionality; protection of fundamental rights (Art. 6(3) TEU).

4 EU Secondary Law

Legal acts adopted by EU institutions on the basis of the founding treaties; cannot contradict primary law. Typology set by **Art. 288 TFEU**.

4.1 Typical & Atypical Acts (Art. 288 TFEU)

Act	Nature	Effect
Regulation	Hard law	Binding in entirety; directly applicable in all member states, no transposition.
Directive	Hard law	Binds as to result ; member states choose form/method (transposition). After the deadline, clear/precise/unconditional provisions have direct effect .
Decision	Hard law	Binding in entirety on those addressed (general or individual); e.g. a Commission decision on specific companies.
Recommendation	Soft law	Not binding; guidance.
Opinion	Soft law	Not binding; guidance.
Atypical acts	—	Communications, resolutions, white papers, green papers — communicate policy / stimulate debate.

4.2 Data-Protection Directives

- **Directive 95/46/EC (1995) — Data Protection Directive**: protection of individuals re processing + free movement of personal data. Adopted by the EC for limited harmonisation. **Repealed and replaced by the GDPR** (contents shifted and expanded).
- **Directive 2006/24/EC — Data Retention Directive** (police enforcement, processing of personal data; reused Dir. 95/46 definitions). **Repealed by the CJEU in *Digital Rights Ireland* (C-293/12 + C-594/12)**.
- **Directive 2002/58/EC — ePrivacy Directive**: processing of personal data *and data in general* in the **electronic communications** sector — confidentiality of communications, cookies, unsolicited marketing, traffic/location data, breach notification. Still in force (not yet replaced); **needs transposition** in each member state. Key definitions: user/subscriber, traffic data, location data, communication.

- **Directive (EU) 2016/680 — Law Enforcement Directive** (data protection in the police/criminal sector): processing for prevention, investigation, detection, prosecution of offences / execution of penalties. Adopted June 2016; **replaced Framework Decision 2008/977/JHA**. Includes data protection by design/default, breach notification to supervisory authorities, mandatory DPOs; decisions based on automated processing must not use sensitive data nor discriminate.

4.3 Data-Protection Regulations

- **Regulation (EU) 2016/679 — GDPR**: comprehensive framework for protection of natural persons re processing + free movement of personal data, in and outside the EU. Detailed in Section 4.4.
- **Regulation (EU) 2018/1725**: protection of natural persons re processing of personal data **by EU institutions**, bodies, offices and agencies + free movement of such data. (Complaints may go to the EDPS.)
- **Framework Decision 2008/977/JHA** — former Justice & Home Affairs instrument on data in police cooperation (national-rules era 1977–2008); **repealed** by Directive 2016/680.

4.4 GDPR — Regulation (EU) 2016/679: key articles

Article	Content
Art. 4	Definitions (personal data, processing, controller, processor, supervisory authority, consent, etc.).
Art. 4(1)	Personal data — any information relating to an identified or identifiable natural person (directly/indirectly: name, ID number, location data, online identifier, or factors of physical/physiological/genetic/mental/economic/cultural/social identity).
Art. 5	Principles of processing : lawfulness, fairness, transparency; purpose limitation; data minimisation; accuracy; storage limitation; integrity & confidentiality; accountability .
Art. 6	Lawful bases for processing (ordinary personal data): consent; contract; legal obligation; vital interests; public-interest task; legitimate interests.
Art. 9	Special categories (“sensitive data”): racial/ethnic origin, political opinions, religious/philosophical beliefs, trade-union membership, genetic & biometric data (for unique ID), health, sex life/sexual orientation. Processing prohibited unless an exception applies (explicit consent, employment/social-security law, legal claims, substantial public interest/public health, non-profit activities, research).
Art. 17	Right to erasure / right to be forgotten; obliges controllers to erase data no longer needed (storage limitation).
Art. 25	Data protection by design and by default — embed TOMs (pseudonymisation, minimisation) from the design phase; default settings process only the minimum necessary.
Art. 28	Controller–processor relationship — must be governed by contract / other legal act.
Art. 51	Supervisory authorities — national independent authorities ensuring application/enforcement; handle complaints; cooperate via the EDPB.

Subjects (GDPR): data subject; controller; (joint controllers); processor; sub-processor; Data Protection Officer (DPO); supervisory authority. **Scope**: material (automated + structured filing systems; household exception) and territorial (**extraterritorial** — non-EU entities offering goods/services to, or monitoring, persons in the Union).

5 2020 European Data Strategy — Legislative Package

Adopted **2020** to build a single market for data (free flow of personal + non-personal data, single market for data) within the Commission’s **Digital Decade (2020–2030)**. Legal basis cited: free economic initiative (CFR, per source Art. 41). The GDPR “paved the way” by ensuring the free flow of personal data under harmonised rules.

Instrument	Year	Aim
Regulation (EU) 2018/1807 (Free Flow of Non-Personal Data)	2018	Free flow of non-personal electronic data; prohibits data-localisation requirements (save public security); fosters a competitive data economy. Mixed datasets: rules apply to the non-personal part.
Data Governance Act (DGA)	2022	Framework for trustworthy data sharing & reuse — public-sector data, data intermediaries, data altruism.
Digital Services Act (DSA)	2022	Safer digital space; platform accountability & transparency ; protects users’ fundamental rights online.
Digital Markets Act (DMA)	2022	Fair competition in digital markets; curbs gate-keepers’ market power; obligations + penalties.
Data Act	2023	Fair access to and use of data generated by connected products/services; cross-sectoral.
AI Act — Regulation (EU) 2024/1689	2024	Risk-based harmonised AI framework; safety, transparency, fundamental rights. Detailed in the companion summary (Section 11).

Note: the DSA, DMA and AI Act do not *strictly* regulate personal-data processing — they govern digitally-natured services/markets that *use* personal data, and can integrate the GDPR.

6 Leading Case Law

Supplementary source of EU law; the Court of Justice fills gaps and develops general principles.

Case	Cite Year	/ Holding
Van Gend en Loos	26/62 [1963]	EU law became an integral part of member states’ legal systems; direct effect — individuals may invoke it before national courts.
Costa v. ENEL	6/64 [1964]	The Community is a New Legal Order for which states limited their sovereign rights; foundation of primacy of EU law.
Schrems v. Facebook	2014	Safe Harbor EU–US agreement declared void; US did not ensure adequate protection of EU citizens’ personal data; led to improved transfer standards.

Case	Cite Year	/	Holding
Google Spain (Costeja Gonzalez)	C-131/12, 2014		Established the Right to be Forgotten ; individuals may request de-listing of outdated/irrelevant links lacking public interest; proportionality of privacy vs. access to information.
Digital Rights Ireland	C-293/12 + C-594/12		Repealed the Data Retention Directive (2006/24/EC).